Macroeconomics I — MPE Insper

Practice list: General Equilibrium, Money and Fiscal Policy — qualifier level (Listas 5–6)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

Block 1 — General equilibrium, hours and taxes

(Kurlat ch. 9; Romer ch. 2 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) *General equilibrium × labor supply.* A two-period economy has a fixed capital stock K and no investment. Households have $u(c) + \psi \ln l$ each period, $u(c) = c^{1-\sigma}/(1-\sigma)$, $\sigma > 0$, $\psi > 0$, and one unit of time, $l = 1 - L$; firms produce $Y = AK^{\alpha}L^{1-\alpha}$ competitively, $0 < \alpha < 1$. At the equilibrium hours $L \in (0,1)$, the elasticity $d\ln L/d\ln A$ is:

A. $1/[L/(1-L) + \alpha]$

B. $(1-\sigma)/[L/(1-L) + \alpha + (1-\alpha)\sigma]$

C. $1/[L/(1-L) + 1]$

D. $(1-\sigma)/[L/(1-L) + 1]$

Answer: B

Step — equilibrium, period by period. With K fixed and no investment, $c = Y$; the firm sets $w = (1-\alpha)Y/L$; the household sets the MRS equal to the wage, $\psi/(1-L) = w\,c^{-\sigma}$. Combining, $\psi L^{\alpha+(1-\alpha)\sigma}/(1-L) = (1-\alpha)(AK^{\alpha})^{1-\sigma}$; the left side rises from 0 to ∞ on $(0,1)$, so L exists and is unique.

Step — log-differentiation. With $d\ln Y = d\ln A + (1-\alpha)\,d\ln L$, the combined FOC gives $[L/(1-L)]\,d\ln L = (1-\sigma)\,d\ln Y - d\ln L$, so $d\ln L/d\ln A$ is B. The denominator is positive: the sign is that of $1-\sigma$, and $\sigma = 1$ is the knife-edge.

Step — distractors. A holds c fixed (partial equilibrium: no income effect from $c = Y$); D treats output as fixed when hours move ($d\ln Y = d\ln A$); C makes both mistakes, holding c fixed and treating output as fixed.

Economic intuition. Higher A raises the wage (substitution) and consumption (income, through $c = Y$); with $\sigma > 1$ the income effect wins.

Ref: Kurlat (2020), §9.1 (pp. 165–168); Lista 5, Q1.

b) (3 points) *General equilibrium × growth.* In Romer's continuous-time Ramsey model with CRRA coefficient $\sigma > 0$, discount rate ρ, population growth $n \geq 0$, technology growth $g > 0$, depreciation $\delta > 0$, $\rho - n - (1-\sigma)g > 0$ and $f(\tilde{k}) = \tilde{k}^{\alpha}$, $0 < \alpha < 1$, the steady-state gross saving rate is:

A. $\alpha(n+g+\delta)/(\rho+\delta+\sigma g)$

B. $\alpha(n+g)/(\rho+\delta+g)$

C. $\alpha(n+g+\delta)/(\rho+\delta+g/\sigma)$

D. $\alpha(n+g+\delta)/(\rho+\delta+g)$

Answer: A

Step — modified golden rule and saving rate. The Euler equation $\dot{\hat{c}}/\hat{c} = [f'(\tilde{k}) - \delta - \rho - \sigma g]/\sigma$ gives $f'(\tilde{k}^*) = \rho + \delta + \sigma g$. Gross investment is $(n+g+\delta)\tilde{k}^*$ and $\tilde{k}/f(\tilde{k}) = \alpha/f'(\tilde{k})$, so $s^* = \alpha(n+g+\delta)/(\rho+\delta+\sigma g)$: A. Also $\alpha - s^* = \alpha[\rho - n - (1-\sigma)g]/(\rho+\delta+\sigma g) > 0$: saving is below the Golden-Rule rate because utility is bounded.

Step — distractors. D applies the log-utility Euler equation ($\sigma = 1$) to CRRA; C reads σ as the intertemporal elasticity ($r = \rho + g/\sigma$); B makes D's mistake and also counts only net investment $(n+g)\tilde{k}$. C and D equal A only at $\sigma = 1$.

Economic intuition. Growth with curvature (σg) raises the required return, so households hold less capital per unit of output.

Ref: Romer (2012), §2.2, §2.5 and Problem 2.6; Kurlat (2020), §9.3.

c) (2 points, T or F) *General equilibrium × fiscal policy.* In the Ramsey model on its balanced growth path, a permanent, unanticipated rise in government purchases per effective worker, financed by lump-sum taxes, lowers consumption one for one on impact and leaves capital unchanged forever.
Answer: T

Step — proof. Lump-sum taxes do not enter the Euler equation, so $\tilde{k}^*$ is unchanged; the resource constraint $\dot{\tilde{k}} = f(\tilde{k}) - \hat{c} - \hat{G} - (n+g+\delta)\tilde{k}$ gives $d\hat{c}^*/d\hat{G} = -1$. Capital starts at the old $\tilde{k}^*$, the new steady-state value; the constant path $(\tilde{k}^*, \hat{c}^* - \Delta\hat{G})$ satisfies both equations and the transversality condition, and the saddle path through a given $\tilde{k}(0)$ is unique.

Economic intuition. A permanent purchase is a permanent resource loss, absorbed entirely by consumption.

Ref: Romer (2012), §2.7.

d) (2 points, T or F) *General equilibrium × taxes.* In the Ramsey model, a permanent tax at rate $\tau_K \in (0,1)$ on capital income, whose revenue is rebated lump-sum, leaves the steady-state capital stock unchanged, because the rebate gives the revenue back.
Answer: F

Step — proof. Households earn $(1-\tau_K)[f'(\tilde{k}) - \delta]$, so the Euler equation becomes $\dot{\hat{c}}/\hat{c} = \{(1-\tau_K)[f'(\tilde{k}) - \delta] - \rho - \sigma g\}/\sigma$ and $f'(\tilde{k}^*) = \delta + (\rho + \sigma g)/(1-\tau_K)$. Implicit differentiation: $d\tilde{k}^*/d\tau_K = (\rho+\sigma g)/[(1-\tau_K)^2 f''(\tilde{k}^*)] < 0$, since $f'' < 0$ and $\rho + \sigma g > n + g \geq 0$. The rebate cancels from the resource constraint: it removes the income effect, not the wedge.

Economic intuition. Saving responds to the marginal after-tax return, which the rebate does not restore.

Ref: Romer (2012), Problem 2.10; Kurlat (2020), Exercise 9.6.

Question 2. (15 points) Discursive. *General equilibrium × labor supply × taxes.* An infinite-horizon economy without growth: households maximize $\int_0^\infty e^{-\rho t}[\ln c + \psi \ln(1-h)]\,dt$, own the capital and rent it to firms; firms produce $Y = K^\alpha h^{1-\alpha}$; capital depreciates at rate δ; labor income is taxed at rate τ and the revenue is rebated lump-sum. Parameters: $0 < \alpha < 1$, $\rho > 0$, $\delta > 0$, $\psi > 0$, $0 \leq \tau < 1$.

a) (3 points) Derive the household's optimality conditions (intertemporal and intratemporal), the firm's conditions and the government budget, and write the equilibrium as a dynamic system in (K, c). Show that hours are a well-defined function $h(K,c)$ and find its elasticities.

Step — household. Budget $\dot{k} = rk + (1-\tau)wh + T - c$. Shifting consumption across a short interval Δ at the return r: $u_c(t) = e^{-\rho\Delta}(1 + r\Delta)u_c(t+\Delta)$; with $u_c = 1/c$ and $\Delta \to 0$, $\dot{c}/c = r - \rho$. Within each instant the MRS equals the after-tax wage: $\psi c/(1-h) = (1-\tau)w$.

Step — firms, government, system. Firms: $r^K = \alpha K^{\alpha-1}h^{1-\alpha}$, $w = (1-\alpha)K^\alpha h^{-\alpha}$; no arbitrage: $r = r^K - \delta$; government: $T = \tau w h$. With T and $r^K K + wh = Y$ (constant returns) in the budget: $\dot{K} = K^\alpha h^{1-\alpha} - c - \delta K$; Euler: $\dot{c} = c(\alpha K^{\alpha-1}h^{1-\alpha} - \rho - \delta)$. Hours solve $\psi c\, h^\alpha/(1-h) = (1-\tau)(1-\alpha)K^\alpha$, whose left side rises from 0 to ∞ on $(0,1)$: $h(K,c)$ exists, is unique and smooth. Log-differentiating, $[\alpha + h/(1-h)]\,d\ln h = \alpha\, d\ln K - d\ln c$:

$$\frac{\partial \ln h}{\partial \ln K} = \alpha\varepsilon > 0, \qquad \frac{\partial \ln h}{\partial \ln c} = -\varepsilon < 0, \qquad \varepsilon \equiv \frac{1}{\alpha + h/(1-h)} = \frac{1-h}{\alpha + (1-\alpha)h}.$$

Economic intuition. ε blends the Frisch elasticity $(1-h)/h$ with diminishing returns to hours.

Ref: Kurlat (2020), §7.4 and §9.3; Romer (2012), §2.2.

Grading rubric.

- Euler and intratemporal conditions: 1.5 pts (quoted, not derived: max. 0.5 pt).
- Firms, no arbitrage, government budget: 0.5 pt.
- System, $h(K, c)$ and its elasticities: 1 pt (unproved signs: 0 for that line).

b) (3 points) Find the steady state. Which ratios and prices does the tax leave unchanged? Find steady-state hours in closed form and show that they are unique.

Step — ratios. $\dot{c} = 0$: $\alpha(K/h)^{\alpha-1} = \rho + \delta$, so $K/h = \bar{k} \equiv [\alpha/(\rho+\delta)]^{1/(1-\alpha)}$, $K/Y = \alpha/(\rho+\delta)$, $r = \rho$, $w = (1-\alpha)\bar{k}^{\alpha}$. $\dot{K} = 0$: $C/Y = 1 - \delta K/Y = \chi/(\rho+\delta)$, $\chi \equiv \rho + (1-\alpha)\delta$. None depends on τ, ψ.

Step — hours. The intratemporal condition with $c = (C/Y)Y$ and $w = (1-\alpha)Y/h$ gives

$$\frac{h^*}{1-h^*} = \Phi \equiv \frac{(1-\tau)(1-\alpha)(\rho+\delta)}{\psi\chi}, \qquad h^* = \frac{\Phi}{1+\Phi} \in (0,1),$$

unique as $h/(1-h)$ rises from 0 to ∞ on $(0,1)$; $K^* = \bar{k}h^*$, $Y^* = \bar{k}^{\alpha}h^*$, $C^* = (C/Y)\,Y^*$.

Economic intuition. Prescott's formula, with C/Y pinned by impatience and depreciation: the tax sets the level of hours, not the proportions.

Ref: Kurlat (2020), §9.3 and Exercise 7.5; Romer (2012), §2.5.

Grading rubric.

- Ratios and prices from the two loci, independent of τ and ψ: 1.5 pts.
- h^* in closed form, unique: 1.5 pts ($C/Y = 1$ assumed: max. half; hours held fixed: max. half).

c) (3 points) The tax rises permanently and unexpectedly from τ_0 to $\tau_1 \in (\tau_0, 1)$. Find the long-run changes in h, K, Y, C, r and w in closed form, and compare the fall in hours with the fall in the net-of-tax rate $1-\tau$.

Step — hours, scale and prices. Let $\nu \equiv (1-\tau_1)/(1-\tau_0) \in (0,1)$. By (b), Φ is proportional to $1-\tau$, so $\Phi_1 = \nu\Phi_0$; with $\Phi_0 = h_0/(1-h_0)$, $q \equiv h_1/h_0 = \nu/[1-(1-\nu)h_0]$. The denominator lies in $(\nu, 1)$, so $\nu < q < 1$: hours fall, proportionally less than $1-\tau$. Locally, $d\ln h^*/d\ln(1-\tau) = 1 - h^* \in (0,1)$. By (b), K, Y and C are fixed multiples of h, so each falls by the same factor q; $K/h = \bar{k}$ is unchanged, so $r = \rho$ and w do not move in the long run.

Economic intuition. The tax shrinks the economy, not its proportions. Hours respond by $1-h^*$, less than the Frisch elasticity $(1-h^*)/h^*$: consumption falls with them and its marginal utility rises.

Ref: Kurlat (2020), §9.3; Romer (2012), §2.5.

Grading rubric.

- q in closed form and $\nu < q < 1$, proved: 1.5 pts (ordering without proof: 0 for that line).
- Same factor for K, Y, C; r and w unchanged, with the reason: 1.5 pts.

d) (3 points) Characterize the transition. Linearize the system in logs at the new steady state, show that it is a saddle, and find the eigenvalues and the half-life in closed form. Find the impact responses of c and h to first order, compare them with the long-run changes, and draw the paths of $\ln K$, $\ln c$ and $\ln h$ (deviations from the old steady state).

Step — log-linear system. Let $\hat{k} = \ln(K/K_1)$, $\hat{c} = \ln(c/c_1)$, $\theta \equiv (1-\alpha)(\rho+\delta)$. Differentiate $\dot{K}/K = Y/K - c/K - \delta$ and $\dot{c}/c = r^K - \rho - \delta$: by (a) and (b), $\partial(Y/K)/\partial(\ln K, \ln c) = [(\rho+\delta)/\alpha](1-\alpha)(\alpha\varepsilon - 1,\ -\varepsilon)$, $\partial(c/K)/\partial(\ln K, \ln c) = (\chi/\alpha)(-1,\ 1)$ and $\partial \ln r^K/\partial(\ln K, \ln c) = -(1-\alpha)(1-\alpha\varepsilon,\ \varepsilon)$, so

$$\begin{pmatrix} \dot{\hat{k}} \\ \dot{\hat{c}} \end{pmatrix} = J \begin{pmatrix} \hat{k} \\ \hat{c} \end{pmatrix}, \qquad J = \begin{pmatrix} \rho + \theta\varepsilon & -(\chi + \theta\varepsilon)/\alpha \\ -\theta(1-\alpha\varepsilon) & -\theta\varepsilon \end{pmatrix} \quad (\varepsilon \text{ at } h_1).$$

Step — saddle. Let $\omega \equiv \alpha + (1-\alpha)h_1 \in (\alpha, 1)$, so $1 + (1-\alpha)\varepsilon = 1/\omega$ and $1 - \alpha\varepsilon = h_1/\omega > 0$. Then $\operatorname{tr} J = \rho$ and

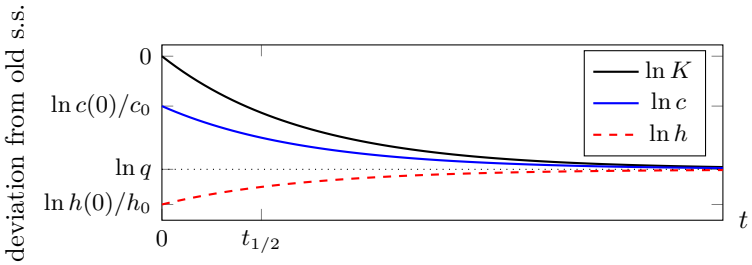
$$\det J = -\frac{\theta\chi}{\alpha\omega} < 0, \qquad \mu_{\pm} = \frac{\rho \pm \sqrt{\rho^2 + 4\theta\chi/(\alpha\omega)}}{2}.$$

Real roots of opposite sign; K is predetermined and c jumps, so the convergent (saddle) path is unique, with speed $\lambda \equiv -\mu_-$ and half-life $t_{1/2} = \ln 2/\lambda$. With $\Lambda \equiv \theta/(\alpha\omega) > (1-\alpha)\delta$, λ is the positive root of $p(x) = x^2 + \rho x - \chi\Lambda$, and $p((1-\alpha)\delta) = \chi[(1-\alpha)\delta - \Lambda] < 0 < \Lambda[\Lambda - (1-\alpha)\delta] = p(\Lambda)$: $(1-\alpha)\delta < \lambda < \Lambda$.

Step — impact and paths. On the stable arm $\hat{c} = \xi\hat{k}$, $\xi = (J_{11}+\lambda)/(-J_{12}) = \alpha(\rho+\theta\varepsilon+\lambda)/(\chi+\theta\varepsilon)$, with $\xi - \alpha = \alpha[\lambda - (1-\alpha)\delta]/(\chi+\theta\varepsilon) > 0$ and $1 - \xi = \alpha(\Lambda - \lambda)/(\chi+\theta\varepsilon) > 0$; hours follow $\hat{h} = \alpha\varepsilon\hat{k} - \varepsilon\hat{c} = -\varepsilon(\xi - \alpha)\hat{k}$. With $\hat{k}(t) = -\ln q\, e^{-\lambda t}$ (K starts at K_0), relative to the old steady state

$$\ln[x(t)/x_0] = \ln q\,(1 - b_x e^{-\lambda t}), \qquad b_K = 1, \quad b_c = \xi, \quad b_h = -\varepsilon(\xi - \alpha).$$

On impact $\ln[c(0)/c_0] = (1-\xi)\ln q \in (\ln q, 0)$: consumption falls by less than in the long run; $\ln[h(0)/h_0] = [1 + \varepsilon(\xi - \alpha)]\ln q < \ln q$: hours overshoot. The figure uses admissible values.



Economic intuition. Capital starts high relative to the new hours, so consumption stays above its new level and the wealth effect pushes hours below theirs; as capital runs down, consumption falls and hours recover.

Ref: Romer (2012), §2.6 (rate of adjustment and slope of the saddle path); Kurlat (2020), §9.3.

Grading rubric.

- Jacobian from the implicit h: 1 pt.
- $\det J < 0$, eigenvalues, saddle, half-life: 0.75 pt (saddle without $\det J < 0$: 0 for that line).
- Impact responses signed through $\alpha < \xi < 1$, proved: 0.5 pt.
- Graph (own line): axes; K from 0, c from $(1-\xi)\ln q$, h below $\ln q$, all tending to $\ln q$: 0.75 pt.

e) (3 points) *Parameter analysis.* Sign the effects of τ, ψ and δ on h^*; on K^*, Y^*, C^*; on K^*/Y^*, C^*/Y^*, r^*, w^*; and on the speed of convergence. Prove each sign, give the limits as $\tau \to 1$, $\psi \to \infty$ and $\psi \to 0$, and explain why the speed changes.

Step — hours and levels. From (b), $d\ln h^* = (1-h^*)\,d\ln\Phi$: $\partial \ln h^*/\partial\tau = -(1-h^*)/(1-\tau) < 0$, $\partial \ln h^*/\partial \ln\psi = -(1-h^*) < 0$ and $\partial \ln h^*/\partial\delta = (1-h^*)[1/(\rho+\delta) - (1-\alpha)/\chi] = (1-h^*)\alpha\rho/[(\rho+\delta)\chi] > 0$. For τ and ψ, K^*, Y^*, C^* move with h^*; the ratios, $r^* = \rho$ and w^* do not move. For δ: $K^*/Y^* = \alpha/(\rho+\delta)$, $C^*/Y^* = 1 - \alpha\delta/(\rho+\delta)$ and $w^* = (1-\alpha)\bar{k}^\alpha$ fall; $\partial \ln K^*/\partial\delta = [\alpha(1-\alpha)(1-h^*)\rho - \chi]/[(1-\alpha)(\rho+\delta)\chi] < 0$ and $\partial \ln Y^*/\partial\delta = \alpha[(1-\alpha)(1-h^*)\rho - \chi]/[(1-\alpha)(\rho+\delta)\chi] < 0$ because $\chi \geq \rho$; so $C^* = (C^*/Y^*)\,Y^*$ falls too.

Step — speed. By (d), $\lambda = (\sqrt{\rho^2 + 4D} - \rho)/2$ rises with $D \equiv \theta\chi/(\alpha\omega)$ at given ρ; τ and ψ lower h^*, hence ω, and so raise λ. For δ: $\partial \ln D/\partial\delta = 1/(\rho+\delta) + (1-\alpha)/\chi - (1-\alpha)h^*(1-h^*)\alpha\rho/[\omega(\rho+\delta)\chi] > 0$, as the last term is below $1/(\rho+\delta)$ ($\alpha \leq \omega$, $\rho \leq \chi$).

Result	τ	ψ	δ
h^*	$-$	$-$	$+$
K^*, Y^*, C^*	$-$	$-$	$-$
K^*/Y^*, C^*/Y^*, w^*	0	0	$-$
$r^* = \rho$	0	0	0
speed λ ($t_{1/2}$)	$+\,(-)$	$+\,(-)$	$+\,(-)$

Step — limits. $\tau \to 1$ or $\psi \to \infty$: $h^* \to 0$ ($K^*, Y^*, C^* \to 0$), $\omega \to \alpha$ and $\lambda \to \bar{\lambda} \equiv (\sqrt{\rho^2 + 4\theta\chi/\alpha^2} - \rho)/2$. $\psi \to 0$: $h^* \to 1$, $\omega \to 1$ and $\lambda \to \underline{\lambda} \equiv (\sqrt{\rho^2 + 4\theta\chi/\alpha} - \rho)/2$, the fixed-labor Ramsey speed; so $\underline{\lambda} < \lambda < \bar{\lambda}$. Knife-edge $\delta = 0$: $C/Y = 1$ and h^* no longer depends on ρ.

Economic intuition. Prices are pinned by $\rho+\delta$, so τ and ψ move only the scale. A lower h^* raises the Frisch elasticity $(1-h)/h$: while consumption is above its new level, hours and output fall more and capital runs down faster; a higher δ makes the interest rate react more to a capital gap.

Ref: Kurlat (2020), §7.4 (Frisch elasticity) and §9.3; Romer (2012), §2.6.

Grading rubric.

- Signs proved: 1.5 pts; table: 0.5 pt; limits: 0.5 pt; reading: 0.5 pt (no proofs or no reading: max. half).

Block 2 — Money, inflation and fiscal policy

(Kurlat chs. 10–11; Romer ch. 11 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Money demand × welfare.* Baumol–Tobin: a household holds its income $Y>0$ in a bond paying the nominal rate $i>0$, spends it evenly over the period and withdraws cash in N equal trips at a cost $F>0$ each (N continuous). The welfare cost of $i>0$, measured as the Bailey area $\int_0^i m(x)\,dx - i\,m(i)$ under the money demand $m(i)$ this problem generates, is:

A. $\sqrt{FYi/4}$

B. $\sqrt{2FYi}$

C. $\sqrt{FYi/8}$

D. $\sqrt{FYi/2}$

Answer: D

Step — money demand. Cash falls linearly from Y/N to 0 between trips, so $m=Y/(2N)$ and the household minimizes $FN+iY/(2N)$; the first-order condition $F=iY/(2N^2)$ (second derivative $iY/N^3>0$) gives $N^*=\sqrt{iY/(2F)}$ and $m(i)=\sqrt{FY/(2i)}$.

Step — Bailey area. The integral converges since $m\propto x^{-1/2}$: $W=\int_0^i\sqrt{FY/(2x)}\,dx - i\,m(i) = \sqrt{2FYi}-\sqrt{FYi/2}=\sqrt{FYi/2}=FN^*$. The first term is the minimized cost $C(i)=FN^*+i\,m(i)$ (envelope theorem: $C'=m$, $C(0^+)=0$); the interest $i\,m$ is a transfer to the issuer of money, so the loss is the cost of trips, Kurlat's shoe-leather cost NF.

Step — distractors. B also counts the forgone interest, a transfer equal to FN^* at the optimum: it is $C(i)=2W$. C takes the triangle $\frac{1}{2}\,i\,m(i)$. A makes C's mistake and also uses average balances Y/N, so $m=\sqrt{FY/i}$ and $\frac{1}{2}\,i\,m=\frac{1}{2}\sqrt{FYi}$.

Ref: Kurlat (2020), ch. 10, §10.4, eqs. (10.4.1)–(10.4.4); ch. 11, §11.4, eq. (11.4.1).

b) (3 points) *Money × expectations.* Cagan with perfect foresight and no bubbles, in logs: $m_t-p_t=-a(p_{t+1}-p_t)$, $a>0$. Money has been constant; at $t=0$ the central bank credibly announces a permanent rise of $\theta>0$ in the log money supply from date $k\geq 1$ on. On impact p_0 rises by:

A. $\left[1-\left(\frac{a}{1+a}\right)^k\right]\theta$

B. $\frac{1}{1+a}\left(\frac{a}{1+a}\right)^k\theta$

C. $\left(\frac{a}{1+a}\right)^k\theta$

D. $\frac{1}{1+a}\left[1-\left(\frac{a}{1+a}\right)^k\right]\theta$

Answer: C

Step — forward solution. Solving for p_t, $p_t=\frac{1}{1+a}m_t+\lambda p_{t+1}$, $\lambda\equiv\frac{a}{1+a}\in(0,1)$; iterating with no bubbles ($\lambda^n p_{t+n}\to 0$), $p_t=\frac{1}{1+a}\sum_{j\geq 0}\lambda^j m_{t+j}$, with weights summing to $\frac{1}{(1+a)(1-\lambda)}=1$.

Step — news. Only m_j, $j\geq k$, rise by θ: $\Delta p_0=\frac{\theta}{1+a}\sum_{j\geq k}\lambda^j=\frac{\theta}{1+a}\,\frac{\lambda^k}{1-\lambda}=\lambda^k\theta$, and p_t rises by $\lambda^{k-t}\theta$ for $t\leq k$: prices climb before the money arrives.

Step — distractors. A is the rise still to come, $(1-\lambda^k)\theta$; B treats the rise as temporary, moving only m_k; D makes both mistakes: the rise still to come under a temporary rise, from impact to its peak $\theta/(1+a)$ at date k.

Ref: Romer (2012), ch. 11, Problem 11.1; Kurlat (2020), ch. 11, Exercise 11.6.

c) (2 points, T or F) With money demand as in b) and money growth long at μ_1, a credible announcement at $t=0$ that money growth will fall permanently to $\mu_2<\mu_1$ after date $T\geq 1$ leaves inflation at μ_1 until money growth falls.

Answer: F

Step — forward-looking prices. By b), $p_t-m_t=\frac{1}{1+a}\sum_{j\geq 0}\lambda^j(m_{t+j}-m_t)$, equal to $a\mu_1$ on the old path; the announcement lowers $m_{t+j}-m_t$ by $(\mu_1-\mu_2)\max\{0,t+j-T\}$, so, by the sums of Question 4(d), $p_t=m_t+a\mu_1-a(\mu_1-\mu_2)\lambda^{T-t}$ for $0\leq t\leq T$. Inflation is $\mu_1-a(\mu_1-\mu_2)\lambda^T$ in the announcement period and $\mu_1-(\mu_1-\mu_2)\lambda^{T-s+1}$ at $1\leq s\leq T$, always below μ_1: it falls at once.

Ref: Romer (2012), ch. 11, Problem 11.1(d); Kurlat (2020), ch. 11, Exercise 11.4.

d) (2 points, T or F) *Money × taxes.* With nominal interest income taxed at $\tau\in(0,1)$ and a constant after-tax real return, the nominal rate rises by more than one point per point of expected inflation, and the before-tax real rate $i-\pi^e$ rises with π^e.

Answer: T

Step — Darby effect. $(1-\tau)i-\pi^e=\bar{r}$ gives $i=(\bar{r}+\pi^e)/(1-\tau)$, so $di/d\pi^e=1/(1-\tau)>1$, and $i-\pi^e=(\bar{r}+\tau\pi^e)/(1-\tau)$ has slope $\tau/(1-\tau)>0$: the tax also falls on the inflation premium. $\tau=0$ restores the one-for-one Fisher effect.

Ref: Kurlat (2020), ch. 11, §11.1, eq. (11.1.1); Romer (2012), ch. 11, §11.3 (inflation and the tax system).

Question 4. (15 points) Discursive. *Money × fiscal policy × expectations: unpleasant arithmetic.* Real money demand and debt, as shares of GDP, are $m=\kappa e^{-a\pi}$ $(\kappa,a>0)$ and $b_{t+1}=(1+r)b_t+d-S_t$, with $r>0$, primary deficit $d\geq 0$ and $b_0>0$; output does not grow, so seigniorage at constant inflation $\pi\geq 0$ is $S=\pi m$. The seigniorage S_0 that would hold debt at b_0 is below the largest steady-state seigniorage (part a). As in the Brazilian debates of the 1980s, the central bank tightens and the Treasury does not adjust: for $T\geq 1$ periods $(t=0,\ldots,T-1)$ seigniorage is $S_1\in(0,S_0)$; from T on, debt is held at b_T.

a) (3 points) Characterize the steady states: the link between debt, deficit and seigniorage; the largest seigniorage; how many inflation rates raise a given S, and which to select and why; and the inflation rate π_0 if debt is stabilized at once, with bounds in terms of S_0 and κ.

Step — budget and Laffer curve. Constant debt requires $S=d+rb$ (deficit plus interest bill), so $S_0=d+rb_0$. At constant π, $S(\pi)=\kappa\pi e^{-a\pi}$, $S'(\pi)=\kappa e^{-a\pi}(1-a\pi)$ and $S''(\pi)=-a\kappa e^{-a\pi}(2-a\pi)$: S rises on $[0,1/a]$ and falls after, $S(0)=0$, $S\to 0$ as $\pi\to\infty$, the unique maximum is $S_{\max}=S(1/a)=\kappa/(ae)$, and S is strictly concave on $[0,2/a)$.

Step — two roots, one selected. For $0<S<S_{\max}$, continuity and strict monotonicity on each side of $1/a$ give exactly one root $\pi_L(S)\in(0,1/a)$ and one $\pi_H(S)>1/a$ (intermediate value theorem); they merge at $S_{\max}$ and none exists above it. On the high branch $\pi_H'=1/S'(\pi_H)<0$, so a smaller financing need would raise inflation; like Romer, we select π_L.

Step — properties of π_L. Inverse-function rule: $\pi_L'(S)=e^{a\pi_L}/[\kappa(1-a\pi_L)]>0$ and $\pi_L''=-S''/S'^3>0$: inflation is increasing and strictly convex in the seigniorage it must raise, and $\pi_L'\to\infty$ as $S\to S_{\max}$. As $e^{-1}<e^{-a\pi}<1$ on $(0,1/a)$, $S/\kappa<\pi_L(S)<eS/\kappa$; so $\pi_0=\pi_L(S_0)$ exists, with $S_0/\kappa<\pi_0<eS_0/\kappa$.

Economic intuition. Inflation taxes real balances; past $\pi=1/a$ the base, shrinking at the semi-elasticity a, outweighs the higher rate.

Ref: Kurlat (2020), ch. 11, §11.3, eq. (11.3.1), and Exercise 11.6; Romer (2012), ch. 11, §11.9, eqs. (11.67)–(11.69).

Grading rubric.

- $S=d+rb$, S', the peak at $1/a$ and $S_{\max}$ derived: 1.5 pts (numerical curve: max. half).
- Two roots proved, low branch justified: 1 pt (roots asserted without monotonicity: 0).

- π_0 with its bounds: 0.5 pt.

b) (3 points) Find debt at T and the seigniorage S_2 needed from T on in closed form, and describe debt during the tight phase. Characterize inflation during (π_1) and after (π_2) the tight phase; when does π_2 exist?

Step — debt. With $S_t = S_1$ the debt equation has the fixed point $\bar{b} = (S_1 - d)/r$ and $b_{t+1} - \bar{b} = (1+r)(b_t - \bar{b})$, so $b_T = \bar{b} + (1+r)^T(b_0 - \bar{b}) = (1+r)^T b_0 + (d - S_1)[(1+r)^T - 1]/r$. As $b_0 - \bar{b} = (S_0 - S_1)/r > 0$, $b_{t+1} - b_t = (1+r)^t(S_0 - S_1) > 0$: debt rises ever faster, without bound if the phase never ends.

Step — seigniorage and inflation later. Freezing debt needs $S_2 = d + rb_T$ (part a); with $d + r\bar{b} = S_1$ and $r(b_0 - \bar{b}) = S_0 - S_1$, $S_2 = S_1 + (1+r)^T(S_0 - S_1)$: the tightness gap compounds at r. Then $\pi_1 = \pi_L(S_1)$ and $\pi_2 = \pi_L(S_2)$, which exists iff $S_2 \le S_{\max}$; both obey the bounds of a).

Economic intuition. Seigniorage not collected today is borrowed at r, and the interest must be paid with seigniorage forever after.

Ref: Kurlat (2020), ch. 11, §11.3, eq. (11.3.1).

Grading rubric.

- b_T by solving the difference equation, and debt rising: 1 pt.
- S_2 in the compounding form: 1 pt (quoted without derivation: max. a third).
- π_1, π_2 on the low branch with the existence condition: 1 pt.

c) (3 points) Compare π_2 with π_0, and compare the r-discounted average inflation $\bar{\pi} \equiv \frac{r}{1+r}\sum_{t\ge 0}(1+r)^{-t}\pi(t)$ under this policy and under immediate stabilization. Find in closed form the length T^* of the tight phase beyond which no steady state exists.

Step — later inflation. By b), $S_2 - S_0 = [(1+r)^T - 1](S_0 - S_1) > 0$ for $r > 0$, $T \ge 1$, $S_1 < S_0$; as π_L is strictly increasing, $\pi_1 < \pi_0 < \pi_2$: lower inflation now is bought with higher inflation forever after.

Step — discounted average. The weights $\frac{r}{1+r}(1+r)^{-t}$ sum to one, those of $t \ge T$ to $\omega \equiv (1+r)^{-T}$, and $(1-\omega)S_1 + \omega S_2 = S_1 + \omega(1+r)^T(S_0 - S_1) = S_0$: the intertemporal budget fixes discounted seigniorage, and the policy only re-times it. By strict convexity of π_L (Jensen),

$$\bar{\pi} = (1-\omega)\pi_L(S_1) + \omega\pi_L(S_2) > \pi_L\big((1-\omega)S_1 + \omega S_2\big) = \pi_0.$$

Step — collapse. A steady state exists iff $S_2 \le S_{\max}$, i.e. $(1+r)^T \le (S_{\max} - S_1)/(S_0 - S_1)$:

$$T^* = \frac{\ln[(S_{\max} - S_1)/(S_0 - S_1)]}{\ln(1+r)} > 0, \qquad S_{\max} = \frac{\kappa}{ae};$$

for $T > T^*$ no inflation rate pays the interest bill.

Economic intuition. The policy only re-times the inflation tax; since each unit of seigniorage costs more inflation than the last, bunching it later raises average inflation.

Ref: Kurlat (2020), ch. 11, §11.3; Romer (2012), ch. 11, §11.9; Sargent and Wallace's (1981) unpleasant monetarist arithmetic.

Grading rubric.

- $S_2 > S_0$ and the ordering via the increasing branch: 1 pt (ordering merely asserted: 0 for this line).
- Unchanged discounted seigniorage and Jensen's inequality: 1 pt.
- T^* derived, with $T^* > 0$: 1 pt.

d) (3 points) Now money demand is forward-looking, $m_t - p_t = -a(p_{t+1} - p_t)$ in logs, with perfect foresight and no bubbles. Money has long grown at μ_1; at $t = 0$ it is credibly announced that $m_t - m_{t-1} = \mu_1$ for $t \le T$ and $\mu_2 > \mu_1$ for $t > T$. Find the price level, its jump on announcement and inflation at every date; characterize inflation before T (bounds, monotonicity, half-life) and the timing of the fall in real balances; read it with $\mu_1 = \pi_1$, $\mu_2 = \pi_2$.

Step — forward solution. By Question 3(b), $p_t - m_t = \frac{1}{1+a}\sum_{j\geq 0}\lambda^j(m_{t+j} - m_t)$, $\lambda \equiv \frac{a}{1+a}$, and two sums do the work: $\frac{1}{1+a}\sum_{j\geq 0} j\lambda^j = \frac{\lambda}{(1+a)(1-\lambda)^2} = a$ and $\frac{1}{1+a}\sum_{j\geq n}(j-n)\lambda^j = a\lambda^n$.

Step — price level. For $t \leq T$, $m_{t+j} - m_t = \mu_1 j + \Delta \max\{0, t+j-T\}$ with $\Delta \equiv \mu_2 - \mu_1 > 0$; for $t \geq T$ it is $\mu_2 j$. Hence $p_t = m_t + a\mu_1 + a\Delta\lambda^{T-t}$ for $0 \leq t \leq T$ and $p_t = m_t + a\mu_2$ for $t \geq T$ (equal at T). The old path was $m_t + a\mu_1$, so p_0 jumps up by $a\Delta\lambda^T$.

Step — inflation. With $a(1-\lambda) = \lambda$: $\pi(s) \equiv p_s - p_{s-1} = \mu_1 + \Delta\lambda^{T-s+1}$ for $1 \leq s \leq T$ and μ_2 after, so $\mu_1 < \pi(s) < \mu_2$, rising in s; the excess $\Delta\lambda^k$, k periods before the switch, halves every $\ln 2/\ln(1+1/a)$ periods. Real balances fall by $a\Delta$ in all, and jump plus excess inflation sum to $a\Delta\lambda^T + \sum_{k=1}^{T}\Delta\lambda^k = a\Delta$: all of it by T. With $\mu_1 = \pi_1$, $\mu_2 = \pi_2$ (b–c), inflation exceeds π_1 throughout the tight phase.

Economic intuition. Expected inflation lowers money demand now, so prices rise before the money does: tight money does not even buy low inflation meanwhile, and a larger a (higher λ) pulls inflation further forward.

Ref: Romer (2012), ch. 11, Problem 11.1; Kurlat (2020), ch. 11, Exercises 11.4 and 11.6.

Grading rubric.

- Forward solution and the two sums: 1 pt.
- Price level, jump and inflation path: 1 pt.
- Bounds, monotonicity, half-life, fall $a\Delta$ by T, read with π_1, π_2: 1 pt (no link to b–c: max. half).

e) (3 points) *Parameter analysis.* Sign, with proofs, the effects of S_1, T, r, d (or b_0), κ, a on S_2, π_2, $\pi_2 - \pi_1$, T^* in a table; give the limits $T \to 0$, $S_1 \to S_0$, $T \to T^*$, $r \to 0$, $a \to 0$ with their reading; draw π_2 against T for two values of S_1.

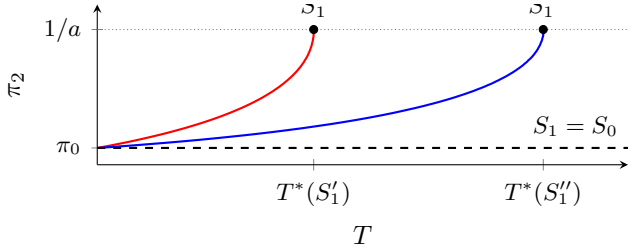
Step — S_2. From $S_2 = S_1 + (1+r)^T(S_0 - S_1)$, $S_0 = d + rb_0$: $\partial_{S_1}S_2 = 1 - (1+r)^T < 0$, $\partial_T S_2 = (1+r)^T \ln(1+r)(S_0 - S_1) > 0$, $\partial_r S_2 = T(1+r)^{T-1}(S_0 - S_1) + (1+r)^T b_0 > 0$, $\partial_d S_2 = (1+r)^T > 0$, $\partial_{b_0}S_2 = r(1+r)^T > 0$; κ, a do not enter.

Step — π_2 and $\pi_2 - \pi_1$. Implicit differentiation of $\kappa\pi e^{-a\pi} = S$ on the low branch ($1 - a\pi > 0$): $\partial_S \pi_L = e^{a\pi}/[\kappa(1-a\pi)] > 0$, $\partial_\kappa \pi_L = -G(\pi)/\kappa$, $\partial_a \pi_L = H(\pi)$, with $G \equiv \pi/(1-a\pi)$ and $H \equiv \pi^2/(1-a\pi)$ positive and increasing on $(0, 1/a)$. So π_2 inherits the signs of S_2, falls with κ, rises with a. Gap: $\partial_{S_1}(\pi_2 - \pi_1) = \pi_L'(S_2)[1-(1+r)^T] - \pi_L'(S_1) < 0$; T, r, d, b_0 move only π_2; as $\pi_2 > \pi_1$, the κ effect $-[G(\pi_2) - G(\pi_1)]/\kappa < 0$ and the a effect $H(\pi_2) - H(\pi_1) > 0$.

Step — T^*. $\partial_{S_1}T^* = \left[\frac{1}{S_0 - S_1} - \frac{1}{S_{\max} - S_1}\right]/\ln(1+r) > 0$ as $S_0 < S_{\max}$; d, b_0, r raise S_0 and lower the positive numerator, and r raises the denominator: all three lower T^*; κ raises $S_{\max}$ and T^*, a lowers both; T does not enter.

Step — limits. $T \to 0$ or $S_1 \to S_0$: $S_2 \to S_0$, $\pi_2 \to \pi_0$ (no tightening, no arithmetic), and $T^* \to \infty$ as $S_1 \to S_0$. $T \to T^*$: $S_2 \to S_{\max}$, $\pi_2 \to 1/a$, $d\pi_2/dT = \pi_L'(S_2)\,\partial_T S_2 \to \infty$: a vertical end where the roots merge. $r \to 0$ ($S_1 < d$): $S_2 \to d$ for all T and $T^* \to \infty$; without compounding, delay is free. $a \to 0$: $S_{\max} \to \infty$, and $\pi_L = (S/\kappa)e^{a\pi_L}$ with $a\pi_L < aeS/\kappa \to 0$ gives $\pi_L \to S/\kappa$ (linear Laffer curve, $\bar{\pi} - \pi_0 \to 0$); $\lambda \to 0$ ends the anticipation of d).

	S_1	T	r	d, b_0	κ	a
S_2	$-$	$+$	$+$	$+$	0	0
π_2	$-$	$+$	$+$	$+$	$-$	$+$
$\pi_2 - \pi_1$	$-$	$+$	$+$	$+$	$-$	$+$
T^*	$+$	0	$-$	$-$	$+$	$-$



Step — graph. Admissible values, $S_1' < S_1'' < S_0$: curves start at π_0, rise, are convex ($\pi_2'' = \pi_L''(\partial_T S_2)^2 + \pi_L'\,\partial_T^2 S_2 > 0$); the tighter one lies above and ends earlier; $S_1 = S_0$ is flat.

Economic intuition. Tighter, longer restraint raises later inflation, more so with higher r, debt or deficit, and elastic demand hastens the collapse. With r above growth (here $r > 0 = g$), restraint without fiscal adjustment postpones inflation and enlarges it.

Ref: Kurlat (2020), ch. 11, §11.3; Romer (2012), ch. 11, §11.9.

Grading rubric.

- Signs proved by implicit differentiation: 1 pt (without proofs or reading: max. half of the item).
- Table, limits with proofs, reading: 1 pt.
- Graph: labelled axes; rising convex curves from π_0 ordered by S_1, vertical at $T^*(S_1)$: 1 pt.

Block 3 — Integrative: growth, interest rates, money and seigniorage

(Kurlat chs. 9–11; Romer ch. 11 • 25 points)

Question 5. (10 points) Objective items.

Items a)–d) use the economy of Question 6 and compare balanced paths, on which i, r and π are constant. A closed economy receives the endowment $Y_t = Y_0(1+g)^t$, $g \geq 0$. The representative household maximizes $\sum_{t\geq 0} \beta^t c_t^{1-\sigma}/(1-\sigma)$ ($\ln c_t$ if $\sigma = 1$), with $\sigma > 0$, $0 < \beta < 1$ and $\beta(1+g)^{1-\sigma} < 1$, and trades one-period real and nominal bonds. Real money demand is $M_t/(P_t Y_t) = \kappa e^{-ai_t}$, $\kappa, a > 0$; the Fisher equation holds exactly, $1 + i = (1+r)(1+\pi)$; policy keeps $i \geq 0$.

a) (3 points) *General equilibrium × money.* Let $\sigma = 1$. The central bank adjusts the money supply so that inflation equals its target $\pi^* > 0$. The equilibrium nominal interest rate is:

A. $i = (1+g)(1+\pi^*)/\beta - 1$

B. $i = \beta(1+\pi^*) - 1$

C. $i = (1+\pi^*)/\beta - 1$

D. $i = \beta(1+g)(1+\pi^*) - 1$

Answer: A

Step — Euler equation and goods clearing. The real-bond first-order condition, $c_t^{-\sigma} = \beta(1+r)c_{t+1}^{-\sigma}$, equates the marginal utility given up today to the discounted marginal utility of the $1+r$ goods received tomorrow. In a closed endowment economy no one can lend in the aggregate, so goods clearing forces $c_t = Y_t$ and $c_{t+1}/c_t = 1+g$:

$$1 + r = \frac{(1+g)^\sigma}{\beta} \overset{\sigma=1}{=} \frac{1+g}{\beta}.$$

The real rate is the one at which the household is content to consume exactly the growing endowment.

Step — Fisher. Dividing the nominal-bond condition $c_t^{-\sigma}/P_t = \beta(1+i)c_{t+1}^{-\sigma}/P_{t+1}$ by the real one gives $1 + i = (1+r)(1+\pi)$; with $\pi = \pi^*$, $1 + i = (1+g)(1+\pi^*)/\beta$. Money is endogenous here: holding $M/(PY)$ constant at that i requires M to grow at $(1+g)(1+\pi^*) - 1$.

Step — distractors. C uses the no-growth rate $1 + r = 1/\beta$, ignoring that rising consumption raises r; D inverts the discount factor, writing $\beta u'(c_t) = (1+r)u'(c_{t+1})$, hence $1 + r = \beta(1+g)$; B makes both mistakes, so $1 + r = \beta$.

Ref: Kurlat (2020), ch. 9, §9.3 (Euler equation); ch. 11, §11.1 (Fisher equation); Romer (2012), ch. 11, §11.1.

b) (3 points) *Seigniorage × interest rates.* A government finances itself by printing money, as Brazil's did before the 1994 Real Plan. Let $g = 0$ and write $r > 0$ for the equilibrium real rate. The money stock grows at a constant rate $\mu > 0$ and seigniorage is $S_t = (M_{t+1} - M_t)/P_t$, so $S/Y = \mu\kappa e^{-ai}$ on a balanced path. The money growth rate that maximizes S/Y is:

A. $\mu^* = 1/a$

B. $\mu^* = 1/[a(1+r)]$

C. $\mu^* = 1/a - r$

D. $\mu^* = 1/[a(1+r)] - r$

Answer: B

Step — rate and base. With $g = 0$, item a) gives $1 + r = 1/\beta$, which does not depend on μ. A constant $M/(PY)$ with constant Y forces $\pi = \mu$, and the exact Fisher equation gives $i = (1 + r)(1 + \mu) - 1$, so $di/d\mu = 1 + r$.

Step — first-order condition. $\ln(S/Y) = \ln \mu + \ln \kappa - a[(1 + r)(1 + \mu) - 1]$, so

$$\frac{d \ln(S/Y)}{d\mu} = \frac{1}{\mu} - a(1 + r) = 0 \quad \Longrightarrow \quad \mu^* = \frac{1}{a(1 + r)}.$$

Step — second-order condition. $d^2 \ln(S/Y)/d\mu^2 = -1/\mu^2 < 0$ for every $\mu > 0$: $\ln(S/Y)$ is strictly concave, so μ^* is the unique global maximum; at μ^*, $d^2(S/Y)/d\mu^2 = -(S/Y)/\mu^{*2} < 0$. Revenue tends to 0 as $\mu \to 0^+$ and as $\mu \to \infty$: the inflation Laffer curve.

Step — distractors. A uses the approximate Fisher equation $i \approx r + \mu$ ($di/d\mu = 1$) and overstates μ^* by the factor $1 + r$; D subtracts r, converting the root as if it were a nominal rate ($\mu = i - r$); C makes both mistakes, subtracting r from A's root.

Economic intuition. At the peak the base κe^{-ai} has elasticity $-a(1 + r)\mu^* = -1$ with respect to μ: a higher tax rate is exactly offset by the shrinking base. The exact Fisher equation makes the base shrink faster, by the factor $1 + r$.

Ref: Kurlat (2020), ch. 11, §11.3 and Exercise 11.6; Romer (2012), ch. 11, §11.9.

c) (2 points, T or F) *General equilibrium × money × welfare.* In this endowment economy a permanent rise in money growth leaves the real interest rate unchanged and therefore has no real effects.

Answer: F

Step — what money growth cannot move. $1 + r = (1 + g)^\sigma/\beta$ and $c_t = Y_t$ (item a) do not involve μ: consumption and the real rate are superneutral.

Step — what it moves. $M/(PY)$ is constant on a balanced path, so $1 + \pi = (1 + \mu)/(1 + g)$ and, with $m \equiv \kappa e^{-ai}$,

$$1 + i = \frac{(1 + g)^{\sigma-1}(1 + \mu)}{\beta}, \qquad \frac{\partial i}{\partial \mu} = \frac{(1 + g)^{\sigma-1}}{\beta} > 0, \qquad \frac{\partial m}{\partial \mu} = -a\, m\, \frac{\partial i}{\partial \mu} < 0.$$

Real balances are a real variable, and they fall.

Step — welfare. The Bailey deadweight loss per unit of output, $w(i)$ of Question 6(d), is

$$w(i) = \int_0^i m(x)\, dx - i\, m(i) = \frac{\kappa}{a}\left[1 - (1 + ai)e^{-ai}\right], \qquad w'(i) = a\kappa\, i\, e^{-ai} = -i\, m'(i) > 0,$$

and $w(i) > 0$ for $i > 0$ because $e^{ai} > 1 + ai$. Higher μ raises i and the loss; only the Friedman rule $i = 0$, that is $1 + \mu = \beta(1 + g)^{1-\sigma}$, removes it. Superneutrality fails once holding money yields services.

Ref: Kurlat (2020), ch. 10, §10.4; ch. 11, §11.2 (neutrality) and §11.4 (cost of inflation, Friedman rule).

d) (2 points, T or F) *Growth × money.* Under a constant money-growth rule, faster trend growth always lowers the nominal interest rate because it lowers inflation.

Answer: F

Step — two channels. From c), $1 + \pi = (1 + \mu)/(1 + g)$ falls with g (growing demand for real balances absorbs part of the money growth), while $1 + r = (1 + g)^\sigma/\beta$ rises with g. From $1 + i = (1 + g)^{\sigma-1}(1 + \mu)/\beta$:

$$\frac{\partial \ln(1 + i)}{\partial \ln(1 + g)} = \underbrace{\sigma}_{\text{real rate}} \quad \underbrace{-\,1}_{\text{inflation}} = \sigma - 1.$$

Step — sign. i rises with g if $\sigma > 1$, does not depend on g if $\sigma = 1$, and falls only if $\sigma < 1$, that is, only when the elasticity of intertemporal substitution $1/\sigma$ exceeds one. The log case already refutes "always".

Economic intuition. Under inflation targeting $\partial \ln(1+i)/\partial \ln(1+g) = \sigma > 0$ for every σ: the money rule passes faster growth into lower inflation one for one, but with $\sigma > 1$ the real rate rises by more.

Ref: Kurlat (2020), ch. 9, §9.3; ch. 11, §11.2 (steady-state inflation); Romer (2012), ch. 11, §11.1.

Question 6. (15 points) Discursive. *Integrative: growth × equilibrium interest rates × money demand × seigniorage × welfare.* An endowment economy produces $Y_t = Y_0(1+g)^t$, $Y_0 > 0$, $g \geq 0$. A representative household maximizes $\sum_{t\geq 0} \beta^t c_t^{1-\sigma}/(1-\sigma)$ (logarithmic at $\sigma = 1$), with $\sigma > 0$ and $\beta \in (0,1)$, and trades one-period real and nominal bonds, both in zero net supply. Real money demand is the reduced form $M_t/(P_t Y_t) = \kappa e^{-a i_t}$, $\kappa, a > 0$, where i_t is the nominal interest rate, and the Fisher relation holds exactly: $1 + i_t = (1 + r_t)(1 + \pi_{t+1})$. The government hands the new money to the household as a lump-sum transfer; with money growing at rate μ and new money valued at today's prices, seigniorage is $S_t/Y_t = \mu\, M_t/(P_t Y_t)$. Assume $\beta(1+g)^{1-\sigma} < 1$ and study steady states (constant r, i, π, μ) with $i \geq 0$. Brazil adopted inflation targeting in 1999 and has lived with high real interest rates; the items ask what this economy says about both. All answers are symbolic.

a) (3 points) Derive the Euler equation and find the equilibrium real interest rate. Show that $\beta(1+g)^{1-\sigma} < 1$ is the condition for household wealth to be finite and say what it implies for r and g. Under a money-growth rule $M_{t+1} = (1+\mu)M_t$, find steady-state inflation, the nominal rate and the admissible range of μ.

Step — household problem. With transfer z_t, real bonds b_t and nominal bonds B_t (in units of money), the budget is $c_t + b_{t+1} + B_{t+1}/P_t = Y_t + z_t + (1+r_{t-1})b_t + (1+i_{t-1})B_t/P_t \equiv x_t$; money demand is a reduced form that does not enter this choice. With $\mathcal{L} = \sum_t \beta^t \{u(c_t) + \lambda_t[x_t - c_t - b_{t+1} - B_{t+1}/P_t]\}$, where λ_t is the marginal utility of goods at t, the FOCs are $u'(c_t) = \lambda_t$, $\lambda_t = \beta(1+r_t)\lambda_{t+1}$ and $\lambda_t/P_t = \beta(1+i_t)\lambda_{t+1}/P_{t+1}$. The first two give the Euler equation $c_t^{-\sigma} = \beta(1+r_t)c_{t+1}^{-\sigma}$; dividing the third by the second gives $1 + i_t = (1+r_t)(1+\pi_{t+1})$, the Fisher relation as no arbitrage between the two bonds.

Step — equilibrium. Bonds are in zero net supply ($b_t = B_t = 0$) and the transfer returns seigniorage, so goods clearing gives $c_t = Y_t$: nobody can lend in the aggregate, and r_t must make the household content to consume its endowment. Putting $c_{t+1}/c_t = 1 + g$ into the Euler equation gives a constant rate,

$$1 + r = \frac{(1+g)^\sigma}{\beta}.$$

Step — finite wealth. Wealth is the present value of the endowment, $\sum_t Y_t/(1+r)^t = Y_0 \sum_t [(1+g)/(1+r)]^t$, finite iff $(1+g)/(1+r) = \beta(1+g)^{1-\sigma} < 1$, that is, iff $r > g$ (the same condition makes $\sum_t \beta^t u(Y_t)$ converge). Also $r > 0$, since $(1+g)^\sigma \geq 1 > \beta$.

Step — money and Fisher. With i constant, real balances $M_t/P_t = \kappa e^{-ai} Y_t$ grow at g, so $1 + \mu = (1+\pi)(1+g)$; then the Fisher relation gives

$$1 + \pi = \frac{1+\mu}{1+g}, \qquad 1 + i = (1+r)(1+\pi) = \frac{(1+g)^{\sigma-1}(1+\mu)}{\beta} \equiv A\,(1+\mu),$$

where $A \equiv (1+r)/(1+g) = 1/[\beta(1+g)^{1-\sigma}] > 1$ by the assumption. Admissibility, $i \geq 0$ (otherwise money dominates bonds), is $1 + \mu \geq 1/A$, a bound below one: the money stock may shrink.

Economic intuition. With a representative agent, growth raises the real rate, not saving: everyone wants to borrow against a richer future, and r rises until nobody does. Money growth is split between output growth and inflation.

Ref: Kurlat (2020), ch. 6, §6.2, eq. (6.2.8); ch. 9, §9.3, eq. (9.3.14); ch. 11, §11.1, eq. (11.1.1), and §11.2, eq. (11.2.3) with unit income elasticity; Romer (2012), ch. 2, §2.5.

Grading rubric.

- Budget, Lagrangian, FOCs and the Euler equation derived: 1 pt (Euler equation quoted: max. a third of this line).
- $c_t = Y_t$ from clearing, $1 + r = (1 + g)^\sigma/\beta$, and finite wealth $\Leftrightarrow r > g$ proved: 1 pt.
- $1 + \pi$ from the money market, $1 + i$ from the exact Fisher relation, admissible μ: 1 pt. Using $r = 1/\beta - 1$ (no growth) or log utility in place of the general case: max. half of the item.

b) (3 points) Compare the money-growth rule with inflation targeting, in which the central bank sets money growth so that $\pi = \pi^*$, with $\pi^* > 0$. For each regime find i and μ, and prove for which σ faster trend growth raises the nominal rate. Decompose the response into the forces at work.

Step — money-growth rule. From (a), $\ln(1 + i) = (\sigma - 1)\ln(1 + g) + \ln(1 + \mu) - \ln\beta$ with μ given, so

$$\frac{\partial i}{\partial g} = (1 + i)\,\frac{\sigma - 1}{1 + g},$$

which has the sign of $\sigma - 1$ because $1 + i > 0$: faster growth raises i iff $\sigma > 1$, leaves it unchanged at $\sigma = 1$ and lowers it iff $\sigma < 1$. Inflation, $1 + \pi = (1 + \mu)/(1 + g)$, falls with g in every case.

Step — inflation targeting. Now $\pi = \pi^*$ and μ is endogenous: the money-market step of (a) gives $1 + \mu = (1 + \pi^*)(1 + g)$, so $\partial\mu/\partial g = 1 + \pi^* > 0$, and the Fisher relation gives

$$1 + i = \frac{(1 + g)^\sigma(1 + \pi^*)}{\beta}, \qquad \frac{\partial i}{\partial g} = (1 + i)\,\frac{\sigma}{1 + g} > 0 \quad \text{for every } \sigma > 0.$$

Step — two forces. Taking logs of $1 + i = (1 + r)(1 + \pi)$,

$$\frac{d\ln(1 + i)}{d\ln(1 + g)} = \underbrace{\frac{d\ln(1 + r)}{d\ln(1 + g)}}_{=\sigma} + \underbrace{\frac{d\ln(1 + \pi)}{d\ln(1 + g)}}_{=-1\text{ (rule), }0\text{ (target)}}.$$

Real-rate force (σ): faster income growth makes households want to borrow; since nobody can, r rises, by more the lower the intertemporal elasticity $1/\sigma$. Inflation force (-1): with μ fixed, real money demand grows with output and absorbs part of the new money. Under the target the central bank supplies that money and the force vanishes, so the target's elasticity exceeds the rule's by exactly one, whatever σ.

Economic intuition. Under a money rule growth is partly disinflationary, and at $\sigma = 1$ the two forces cancel. Under a target only the real-rate force is left: a central bank that targets inflation must raise i when trend growth accelerates.

Ref: Kurlat (2020), ch. 11, §11.2 and Exercise 11.5 (inflation targeting); Romer (2012), ch. 11, §11.1.

Grading rubric.

- Rule: $\partial i/\partial g$ in closed form and its sign $\Leftrightarrow \sigma \gtrless 1$ proved: 1 pt (sign asserted without the derivative: 0 for this line).
- Target: μ endogenous, i, and $\partial i/\partial g > 0$ for every σ: 1 pt.
- Decomposition with both elasticities and their reading: 1 pt. Only one regime analysed: max. half of the item.

c) (3 points) Under the money-growth rule, find the money growth rate μ^* that maximizes seigniorage, check the second-order condition, prove that μ^* is the unique admissible maximizer, find the maximum revenue $s_{\max}$ in closed form, and determine how trend growth moves the peak.

Step — revenue. By (a), $i = A(1 + \mu) - 1$, so seigniorage is a tax rate μ times a base κe^{-ai} that shrinks with μ:

$$s(\mu) \equiv \frac{S}{Y} = \kappa\,\mu\,e^{-a[A(1+\mu)-1]}.$$

Step — first-order condition. $s'(\mu) = \kappa e^{-ai}(1 - aA\mu)$ vanishes only at

$$\mu^* = \frac{1}{aA} = \frac{\beta(1 + g)^{1-\sigma}}{a} > 0,$$

where the elasticity of the base with respect to μ, $-aA\mu$, equals -1.

Step — second order and uniqueness. $s''(\mu) = \kappa a A e^{-ai}(aA\mu - 2)$, so $s''(\mu^*) = -\kappa a A e^{-ai^*} < 0$. Globally, s' has the sign of $1 - aA\mu$, positive below μ^* and negative above, so μ^* is the unique maximizer. It is admissible: $i^* = A(1+\mu^*) - 1 = (A-1) + 1/a > 0$ because $A > 1$.

Step — maximum revenue. Substituting μ^* and i^*,

$$s_{\max} = \kappa \mu^* e^{-ai^*} = \frac{\kappa}{aA}\, e^{-1-a(A-1)} = \frac{\kappa\beta}{a(1+g)^{\sigma-1}} \exp\Big\{ -1 - a\Big[\frac{(1+g)^{\sigma-1}}{\beta} - 1\Big]\Big\}.$$

Step — growth and the peak. $\partial A/\partial g = (\sigma-1)A/(1+g)$. Hence $d\ln\mu^*/d\ln(1+g) = 1-\sigma$ and $\partial i^*/\partial g = (\sigma-1)A/(1+g)$; and since $\ln s_{\max} = \ln\kappa - \ln a - \ln A - 1 - a(A-1)$ has $\partial \ln s_{\max}/\partial A = -(1/A + a) < 0$,

$$\frac{\partial \ln s_{\max}}{\partial g} = -\frac{(\sigma-1)(1+aA)}{1+g}.$$

So μ^* and $s_{\max}$ fall with g iff $\sigma > 1$ (rise iff $\sigma < 1$, constant at $\sigma = 1$), while i^* moves the opposite way. Inflation at the peak, $1 + \pi^L = (1+\mu^*)/(1+g) = \frac{1}{1+g} + \frac{\beta}{a(1+g)^{\sigma}}$, falls with g for every $\sigma > 0$, both terms being decreasing.

Economic intuition. The base is real balances and the tax rate on holding them is i. With $\sigma > 1$ growth raises i at any μ, so the base is smaller and the peak comes earlier and lower; with $\sigma < 1$ the inflation force dominates and the peak moves out.

Ref: Kurlat (2020), ch. 11, §11.3, Exercises 11.3 (seignorage with growth) and 11.6 (Cagan money demand); Romer (2012), ch. 11, §11.9.

Grading rubric.

- $s(\mu)$ with $i(\mu)$ from (a), FOC and μ^*: 1 pt. Using $i = r + \mu$ as if $g = 0$: max. half of the item.
- $s''(\mu^*) < 0$, global uniqueness and $i^* > 0$: 1 pt (SOC asserted without s'': 0 for this line).
- $s_{\max}$ in closed form and each effect of g proved: 1 pt.

d) (3 points) Measure the welfare cost of a steady state with nominal rate i, relative to the Friedman rule $i = 0$, by the Bailey area $W(i)$: the area under the inverse money-demand curve between the balances held at i and at $i = 0$. Find $w(i) \equiv W(i)/Y$ in closed form, prove that it rises with i and characterize it for small and large i. Find the inflation and money growth rates that implement the Friedman rule. Under the target π^*, with nominal rate i_T, find the cost per unit of seigniorage, average and marginal. Draw the demand curve and the cost at the target.

Step — inverse demand. With $m \equiv M/(PY)$, $m(i) = \kappa e^{-ai}$ inverts to $i(m) = \frac{1}{a}\ln\frac{\kappa}{m}$ on $(0, \kappa]$: the height is the marginal value of one more unit of real balances. Balances fall from κ (at $i = 0$) to $m(i)$.

Step — Bailey area. Substituting $m = \kappa e^{-ax}$, $dm = -a\kappa e^{-ax}dx$, and integrating by parts,

$$w(i) = \int_{m(i)}^{\kappa} \frac{1}{a}\ln\frac{\kappa}{m}\, dm = \int_0^i a\kappa\, x\, e^{-ax}\, dx = \kappa\Big[\frac{1 - e^{-ai}}{a} - i\, e^{-ai}\Big],$$

which equals $\int_0^i m(x)\,dx - i\,m(i)$: the surplus lost by holders net of the transfer $i\,m(i)$ to the issuer.

Step — shape. $w'(i) = \kappa a\, i\, e^{-ai} > 0$ for $i > 0$ and $w'(0) = 0$: the cost is second order at the Friedman rule. $w''(i) = \kappa a e^{-ai}(1 - ai)$: convex below $i = 1/a$, concave above. Since $e^{-ax} \le 1$ in the integrand, $w(i) \le \kappa a i^2/2$ (the Harberger triangle), with ratio tending to one as $ai \to 0$. Writing $w = \frac{\kappa}{a}[1 - (1+ai)e^{-ai}]$, the cost is bounded by κ/a and tends to it as $i \to \infty$.

Step — Friedman rule. $i = 0$ requires $(1+r)(1+\pi_F) = 1$:

$$1 + \pi_F = \frac{\beta}{(1+g)^{\sigma}} < 1, \qquad 1 + \mu_F = (1+\pi_F)(1+g) = \beta(1+g)^{1-\sigma} < 1.$$

Deflation at the rate $r/(1+r)$ and a shrinking money stock although output grows; seigniorage $\kappa\mu_F < 0$ must be raised by other taxes.

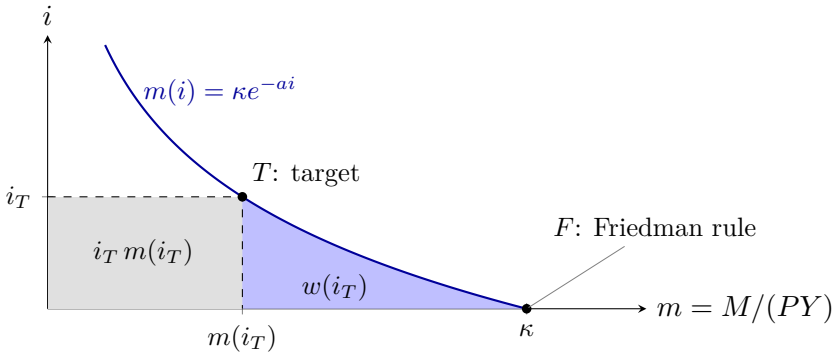
Step — cost per unit of seigniorage. At the target, $1 + i_T = (1+g)^\sigma(1+\pi^*)/\beta$, $\mu_T = (1+\pi^*)(1+g) - 1 > 0$ and $s_T = \kappa\mu_T e^{-ai_T}$. Average cost:

$$\frac{w(i_T)}{s_T} = \frac{e^{ai_T} - 1 - ai_T}{a\,\mu_T} > 0.$$

Marginal cost, raising π^* ($di_T = (1+r)\,d\pi^*$, $d\mu_T = (1+g)\,d\pi^*$):

$$\frac{dw}{ds} = \frac{w'(i_T)(1+r)}{\kappa e^{-ai_T}[(1+g) - a\mu_T(1+r)]} = \frac{aA\,i_T}{1 - aA\mu_T},$$

positive iff $\mu_T < \mu^*$ and unbounded as $\mu_T \uparrow \mu^*$.



Economic intuition. The cost depends on the nominal rate, not on inflation: with a high r, even a low π^* means a sizeable i_T and a sizeable triangle, and the Friedman rule removes it only by deflating at about r.

Ref: Kurlat (2020), ch. 10, §10.4; ch. 11, §11.4 (the Friedman rule); Romer (2012), ch. 11, §11.3 ("The Costs of Inflation").

Grading rubric.

- $w(i)$ derived as the area under the inverse demand (or as $\int_0^i m - i\,m(i)$), in closed form: 1 pt (formula quoted: max. a third of this line).
- $w' > 0$, $w'(0) = 0$, bounds, limits, π_F, μ_F: 0.5 pt. Average and marginal cost per unit of seigniorage: 0.5 pt.
- Graph (scored separately): axes m, i; a decreasing convex curve reaching $(\kappa, 0)$; the area between $m(i_T)$ and κ shaded apart from the transfer rectangle; Friedman point marked: 1 pt (graph contradicting the algebra: 0).

e) (3 points) *Parameter analysis.* Sign the effects of σ, g, β, a and κ on r, on i under each regime, on μ^* and $s_{\max}$, and on the cost at the target, $w_T \equiv w(i_T)$, proving each sign or giving the condition that decides it. Study $\sigma \to 0$, $\sigma = 1$, $g \to 0$ and $a \to 0$, and read the results for Brazil.

Step — log forms. $\ln(1+r) = \sigma\ln(1+g) - \ln\beta$; $\ln(1+i) = (\sigma-1)\ln(1+g) + \ln(1+\mu) - \ln\beta$ (rule) and $\sigma\ln(1+g) + \ln(1+\pi^*) - \ln\beta$ (target); $\ln\mu^* = \ln\beta + (1-\sigma)\ln(1+g) - \ln a$; $\ln s_{\max} = \ln\kappa - \ln a - \ln A - 1 - a(A-1)$ with $\ln A = (\sigma-1)\ln(1+g) - \ln\beta$. So σ enters through $\pm\ln(1+g)$, positive if $g > 0$ and zero at $g = 0$; g through $\sigma/(1+g)$ or $\pm(\sigma-1)/(1+g)$; β through $\mp 1/\beta$. For $s_{\max}$, chain through $\partial\ln s_{\max}/\partial A = -(1/A + a) < 0$; directly, $\partial\ln s_{\max}/\partial a = -1/a - (A-1) < 0$. Neither a nor κ enters r or i.

Step — cost at the target. w_T depends on (σ, g, β) only through i_T, and $w' > 0$, so it inherits the signs of i under the target; it is proportional to κ. For a: $w_T = \kappa h(ai_T)/a$ with $h(x) = 1 - (1+x)e^{-x}$, so $\partial w_T/\partial a = \kappa q(ai_T)/a^2$, where $q(x) \equiv xh'(x) - h(x) = (1 + x + x^2)e^{-x} - 1$. Since $q(0) = 0$, $q'(x) = x(1-x)e^{-x}$, $q(1) = 3/e - 1 > 0$, $q(2) = 7/e^2 - 1 < 0$ and $q \to -1$, q is positive on $(0, \hat{x})$ and negative beyond its unique positive root $\hat{x} \in (1, 2)$: a more interest-elastic demand raises the cost iff $ai_T < \hat{x}$.

	σ	g	β	a	κ
r	$+^\dagger$	$+$	$-$	0	0
i, money rule	$+^\dagger$	$\operatorname{sgn}(\sigma-1)$	$-$	0	0
i, target	$+^\dagger$	$+$	$-$	0	0
μ^*	$-^\dagger$	$\operatorname{sgn}(1-\sigma)$	$+$	$-$	0
$s_{\max}$	$-^\dagger$	$\operatorname{sgn}(1-\sigma)$	$+$	$-$	$+$
w_T	$+^\dagger$	$+$	$-$	$+$ iff $ai_T < \hat{x}$	$+$

† 0 when $g = 0$.

Step — limits and knife-edges. $\sigma \to 0$ (linear utility): $1+r \to 1/\beta$ and no longer responds to growth; admissibility becomes $\beta(1+g) < 1$; under the rule $1+i \to (1+\mu)/[\beta(1+g)]$ falls with g (only the inflation force is left) and $\mu^* \to \beta(1+g)/a$ rises with g. $\sigma = 1$ (log): $1+i = (1+\mu)/\beta$, $\mu^* = \beta/a$ and $s_{\max}$ do not depend on g, because the two forces cancel; only under the target does i rise with g. $g \to 0$: $1+r \to 1/\beta$, σ drops out of every result (its column becomes 0), the regimes coincide when $\pi^* = \mu$, $1+\pi_F \to \beta$, $\mu^* \to \beta/a$ and $s_{\max} \to \frac{\kappa\beta}{a}e^{-1-a(1/\beta-1)}$. $a \to 0$: $w_T \le \kappa a i_T^2/2 \to 0$ and $\mu^* \to \infty$, so an interest-inelastic demand has no distortion and no interior Laffer peak.

Step — reading for Brazil. (i) High real rates: here r is high only through impatience (low β) or fast growth priced with a low elasticity (high σ and g); with modest trend growth, a persistently high r points to scarce saving (low β) or to forces outside the model, such as risk premia. (ii) Inflation targeting: the policy rate tracks r one for one ($d\ln(1+i)/d\ln(1+r) = 1$), and meeting π^* costs $w(i_T)$ with $1+i_T = (1+r)(1+\pi^*)$: a high real rate makes even a low target costly, and the Friedman rule would demand deflation near r. (iii) Under the target seigniorage is a by-product, $\mu_T = (1+\pi^*)(1+g)-1$; leaning on the inflation tax moves μ_T toward μ^*, where the marginal cost explodes.

Economic intuition. Growth reaches i through two channels of opposite sign and the regime decides whether the inflation channel operates; welfare depends on i, hence on r, not on π alone.

Ref: Kurlat (2020), ch. 9, §9.3; ch. 11, §§11.2–11.4; Romer (2012), ch. 2, §2.5; ch. 11, §11.9.

Grading rubric.

- Table complete and consistent with (a)–(d): 1 pt.
- Each sign proved (log derivatives; $q(x)$ and $\hat{x}$ for a on w_T): 1 pt (a sign without proof: 0 for that entry).
- Limits $\sigma \to 0$, $\sigma = 1$, $g \to 0$, $a \to 0$ and the reading for Brazil: 1 pt. Signs without proofs or without the reading: max. half of the item.